# Intonation

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**Abstract**

Intonation, the grammatical use of voice pitch, is essential for acquisition, speech planning and processing, while it also plays a key role in encoding information structure and conversational implicatures. Thus, a better understanding of its phonological structure and phonetic realization is relevant for research in all these areas. Here, we provide an overview of the essential tenets of the widely adopted autosegmental-metrical (AM) theory of intonational phonology and review the empirical evidence for them. We discuss phonetic variability, which is often seen as a thorny problem in intonation research, and present a number of recent solutions to address it, focusing on methods we adopt in our own research, and also covering the insights that combining research on intonation phonetics with the study of pragmatics has contributed. We conclude with an overview of where these recent findings lead with respect to our understanding of intonation phonetics, phonology, and typology.

## 1. Introduction

The term *intonation* refers to postlexical, utterance-level changes in pitch which serve to chunk utterances into smaller phrases and highlight parts of the message within those phrases in ways that are pragmatically relevant (Ladd, 2001). Thus, intonation is used to express grammatical functions, such as information structure and sentence modality, but also to encode implicatures (Hirschberg, 2004). For instance, if in response to *did John eat the marmalade?* one replies with *Amelia ate the marmalade* using the tune in Figure 1a, the utterance is likely to be treated as a correction; if, on the other hand, the speaker uses the tune in Figure 1b, which shows a more considerable dip and later rise on *Amelia* than the tune in 1a, the answer is more likely to be additionally treated as criticism of Amelia's eating habits.

Following the definition given above, the present paper does not cover lexical tonal phenomena, that is, tone and lexical pitch accent. Additionally, we maintain a distinction between the phonological phenomenon of intonation *per se* and fundamental frequency (henceforth, F0), the main phonetic exponent of intonation which arises from the rate at which the vocal folds vibrate and is perceived as pitch. Finally, our discussion excludes paralinguistic and indexical uses of F0, such as raising pitch to indicate emotion or using greater pitch dynamism (informally, the rate of occurrence of pitch rises and falls) to perform gender.

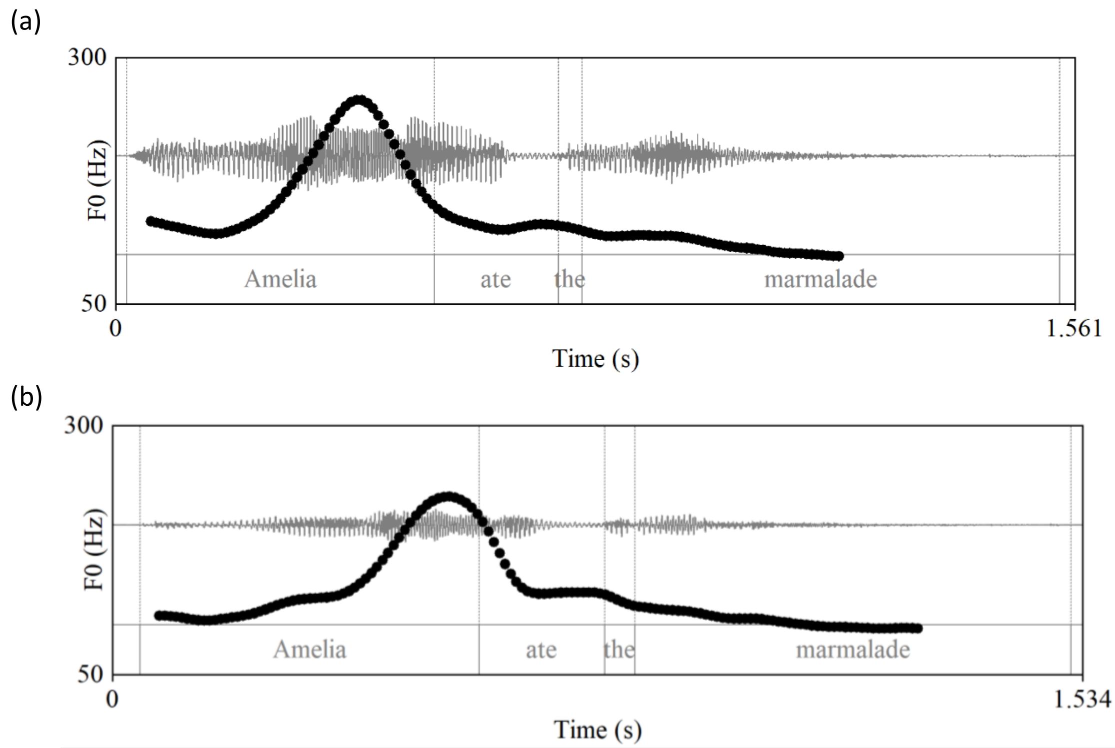


Figure 1: Waveforms and pitch tracks of two renditions of *Amelia ate the marmalade*, as produced by a Californian male speaker of English.

**2. AM phonology and phonetics**

Intonation is often confounded with F0 and considered part of the *suprasegmental* properties of speech (e.g. Ladd, 2008, ch.1). The distinction between segments and suprasegmentals is a layering metaphor that portrays vowels and consonants are discrete and unaffected by suprasegmentals, which are treated as optional icing-on-the-cake layers. As we show, the interaction between segments and intonation is pervasive refuting this ubiquitous metaphor.

Additionally, intonation is often seen as not quite part of the linguistic system or as being primarily driven by biological considerations (Gussenhoven, 2004, ch. 5), an approach dubbed “universalist” by Ladd (2001). Though cross-linguistic research on intonation is still relatively limited, compared to the study of other aspects of speech organization, it is clear that intonation systems are language specific and thus best viewed as a component of a language’s phonology.

The specific understanding on intonation presented here is based on the autosegmental-metrical theory of intonational phonology (AM). AM is based on Pierrehumbert (1980) and widely adopted for both descriptive and experimental research in numerous languages (see Ladd, 2008, and Arvaniti, 2022a, for comprehensive and critical overviews of AM; see Jun, 2005a, 2014a, and Gussenhoven & Chen, 2020, for descriptive studies). Here we provide the essentials of AM and review recent developments.

In AM, intonation is seen not as a "suprasegmental" aspect of the speech signal but as a component of a language's post-lexical phonology. Specifically, the AM conceptualization of intonation relies on two phonological components: the autosegmental string, which represents the melody, and the metrical structure, which represents phrasing and the relative strength of constituents. The autosegmental string is a sequence of low and high tones, represented as L and H, respectively. Tones are *autosegments*, i.e., discrete phonological elements that are independent of segments and associate with structural positions, typically heads and boundaries of metrical constituents. The link between the two components is an essential tenet of AM, as the function of tones is interpreted only in relation to the metrical structure.

Tones that associate with metrical heads are referred to as *pitch accents*, and are distinguished from other tones by means of the star (*) notation, viz. T*. In AM, pitch accents *cue* prominence: their presence indicates that the accented word is metrically strong, though the accent itself does not render the word prominent or "lend" it prominence (Ladd 2008: 54). In AM, pitch accents can be monotonal, e.g., L*, but may also consist of more than one tone, e.g., L+H*. In bitonal accents, one tone is usually starred to indicate it is the strongest component of the accent (for discussions of starredness, see Arvaniti et al., 2000; Gussenhoven, 2002).

*Edge tones* associate with constituent boundaries and cue their presence. Two or three types of edge tones are typically posited, depending on the language. For instance, English, German, and Greek are analyzed as having two phrasal levels, the intermediate phrase or ip and the intonational phrase or IP (Beckman & Pierrehumbert, 1986, Grice et al., 2005, and Arvaniti & Baltazani, 2005, respectively). Phrasing in Japanese and Korean is analyzed as including the *accentual phrase* or AP, which typically contains a content word and its clitics (Pierrehumbert & Beckman, 1988, on Japanese; Jeon, 2015, on Korean). The notation of edge tones varies, depending on the model, the most prevalent being Ta, T- and T%, for AP, ip and IP, respectively. In systems that recognize edge tones associating with the ip and IP, the tones are known as *phrase accents* and *boundary tones,* respectively. Like pitch accents, edge tones may be bi- or multi-tonal; e.g., Jun (2005b) posits LH% and HL% boundary tones for Korean.

An illustration of the above is the structure in Figure 2a. It includes a simplified prosodic tree (based on Pierrehumbert & Beckman, 1988), involving three prosodic words (PrWd or ω) grouped into two ips and forming one IP. Each ip has one pitch accent associated with its' strongest metrical constituent.[1] The illustration also includes the autosegmental string that represents the tune, and the ways in which it associates with prosodic tree nodes. The horizontal lines indicate the separation of the tune and tree.

Phonetically, AM posits that the autosegments making up a melody are realized as sequences of *tonal targets* (or *turning points*), usually F0 minima and maxima. Targets are characterized by their *scaling*

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[1] We note that this is one of several possible groupings, the exact form of which depends on factors such as constituent length, speaking rate and style; for instance, a more deliberate rendition may include an accent on *ate* as well, while a more casual one may group the entire utterance into one ip.

(absolute or relative F0 value) and *alignment* (absolute or relative timing with respect to specific segmental landmarks, such as the onset of a stressed vowel). The tonal targets of pitch accents align with the stressed syllable of the word the accent associates with, while edge tones are realized on sonorant segments adjacent to the relevant boundary (e.g., Silverman & Pierrehumbert, 1990, on American English; Arvaniti et al., 1998, on Greek; Ladd et al., 1999, on British English; D'Imperio, 2001, on Neapolitan Italian). The course of F0 beyond these targets is derived by interpolation, which is largely assumed to be linear. This conceptualization implies that neither the phonological representations nor the phonetic module of AM are meant to fully describe or interpret the F0 contour. In short, AM representations are not phonetic transcriptions of tunes (for discussions, see Arvaniti & Ladd, 2009, Arvaniti, 2016, Arvaniti et al., 2024). The relationship between the phonological representation and phonetics of intonation posited by AM is illustrated in the tune in Figure 2b which reflects the phonological representation in Figure 2a.[2]

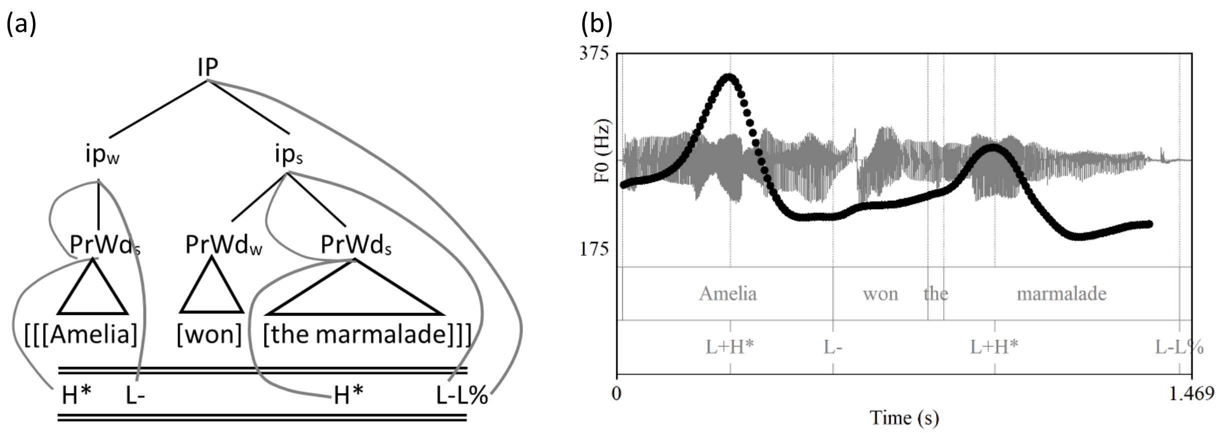


Figure 2. In (a), prosodic tree and autosegmental representation of the tune illustrated in (b), produced by a female speaker of Standard Southern British English. In (a), "w" and "s" indicate whether the constituent is metrically weak or strong within its domain; accents percolate from ips to prosodic words.

To summarize, there are three critical tenets of AM that distinguish it from other intonation models. First, intonation is part of phonology. This is an essential difference, as it means that abstract phonological categories mediate the link between phonetic substance and meaning. For instance, F0 is not directly linked to information structure; rather, it reflects phonological structure, such as the presence of a specific pitch accent (Cangemi & Grice, 2016; Arvaniti, 2019; Arvaniti et al., 2024). An additional consequence of assuming that intonation is part of phonology, is that phonetic realizations are not expected to be invariable but subject to linguistic, contextual, stylistic, dialectal, and sociolinguistic variation. We address this point in the remainder of the paper.

[2] The slight rise at the end of the tune is a speaker-specific feature related to exhalation and does not reflect an audible rise in pitch.

Second, the separation of the melody from metrical strength relationships disentangles stress from intonation and refutes ubiquitous assertions that pitch is a correlate of stress (an idea that, as Ladd & Arvaniti, 2023, show, is based on a misinterpretation of Fry 1955, 1958). Although stressed syllables often have high pitch, particularly in statements, they may also have low pitch, either because they carry a L* accent or because they are unaccented. For example, if *Amelia won the marmalade* is produced as a correction, the stressed syllable of *Amelia* has the highest pitch (see Figure 3a; cf. Figure 1a); however, if the same utterance is produced as a question indicating surprise, the stressed syllable of *Amelia* has the lowest pitch in the utterance (Figure 3b). Moreover, the stress pattern of *marmalade* remains the same in both utterances and is detectable thanks to changes in vowel quality (Beckman & Edwards, 1994), regardless of the marked changes in pitch. In AM, these changes are not an issue with respect to stress, as F0 is determined by the tune, not stress.

Finally, in AM, unlike other models, melodies and their meaning are both compositional. Overall, treating the form and meaning of intonation as compositional has the advantage of providing significant generalizations about melodies, by linking superficially different F0 contours to the same abstract representation and meaning, while also providing an explanation for the differences (for extensive discussion and illustrations, see Arvaniti, 2022a, and Arvaniti et al., 2024). For example, in Figure 4, *jam* and *the marmalade* are produced with the rising tune known as *uptalk*, used in many varieties of English for new information statements (Ritchart & Arvaniti, 2014). In *jam*, the rise starts half way through the vowel, while in *the marmalade*, it is localized on the last, unaccented vowel; additionally, the accent on *marmalade* is much more clearly realized than on *jam*. We provide an explanation of these differences in our discussion of tonal crowding in §3, but independently of them, both pitch contours can be represented as H* L-H%. This representation conveys the information that phonologically both utterances are produced with the same tune but also that they have the same pragmatic meaning. We address the issue of meaning in §5.

(a)

(b)

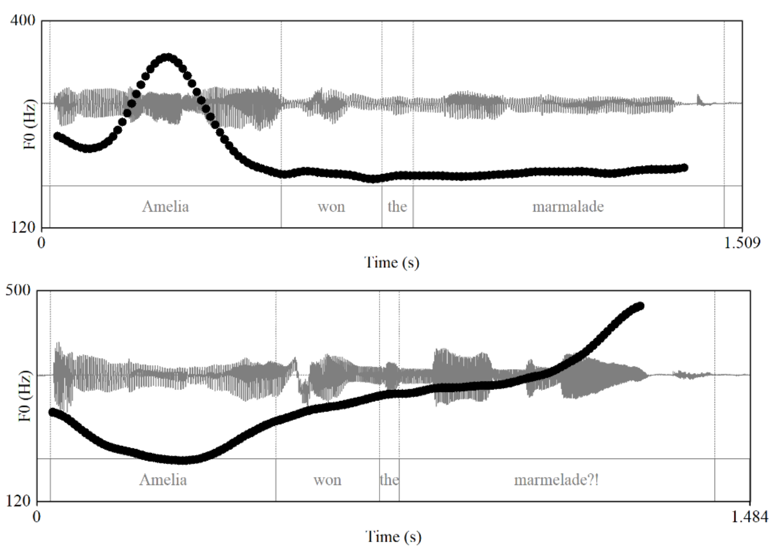


Figure 3. Waveforms and pitch tracks of *Amelia won the marmalade* as produced by a female speaker of Standard Southern British English, as a correction regarding who won the marmalade (a), and as a question expressing surprise (b).

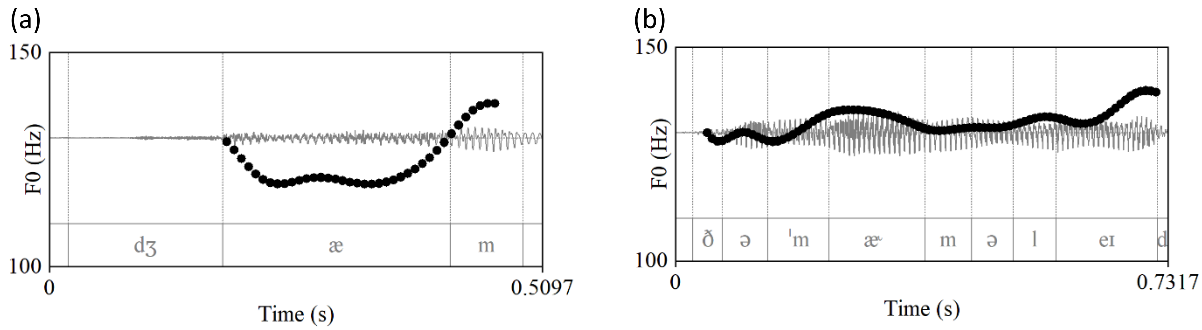


Figure 4. Waveforms and pitch tracks of *jam* (panel a) and *the marmalade* (panel b) as produced by a Californian male speaker of English, using uptalk.

The above understanding of AM phonology has been supported by numerous analyses of the intonation of typologically diverse languages. These include not only languages that have stress, that is, are typologically similar to English, such as Greek (Arvaniti & Baltazani, 2005) and Portuguese (Frota, 2012), but also languages without stress, such as Korean (Jun, 2005b), languages with lexical pitch accent, such as Japanese (Pierrehumbert & Beckman, 1988, Venditti, 2005), and languages with lexical tone, such as Cantonese (Wong et al., 2005).

## 3. Intonation and phonetic variability

A great deal of traditional intonation research includes prosodic annotation as its first step. As illustrated in Figure 2b, annotation involves the categorization of tonal events, which are then acoustically measured and statistically analyzed.[3]

Prosodic annotation is a challenging task in which tonal targets and their alignment with segmental landmarks have played a significant role. Many studies have explored the hypothesis that tonal targets do not simply co-occur but are “anchored” to (i.e., strictly timed with) specific segmental landmarks (Arvaniti et al., 1998, on Greek; Ladd et al., 1999, on English; Ladd et al., 2000, on Dutch). Segmental anchoring has been investigated in numerous languages, including, among others, Neapolitan Italian (D’Imperio, 2001), Basque (Elordieta & Hualde, 2003), Irish (Dalton & Ní Chasaide, 2005), Chickasaw (Gordon, 2008), and Catalan (Prieto, 2009). Research has also uncovered links akin to anchoring between F0 and articulatory gestures (see Byrd & Krivokapic, 2021, for an overview).

In much of this research, tonal alignment stability has been used as a criterion for determining whether putative targets are reflexes of underlying tones. However, as research expanded, empirical evidence for the variability of tonal alignment accumulated. Studies have shown that alignment varies, based on

[3] On annotation, see, Jun (2022) for an overview of ToBI (Tones and Break Indices), the well-known annotation system based on AM; for a critical review of assumptions behind ToBI, see Ladd (2022).

several factors. Many are linguistic in origin and collectively referred to as *lawful variability* by Arvaniti & Ladd (2009). Sources of such variability include speaking rate (Fougeron & Jun, 1998, on French), syllable structure (D'Imperio, 2000, on Neapolitan Italian), phonological weight (Schepman et al., 2006, on Dutch), the identity of edge tones (Cangemi & Grice, 2016, on Neapolitan Italian), segment identity (Grice et al., 2015, on Tashlhiyt Berber), and dialectal differences (Atterer & Ladd, 2004, on German; Burdin et al., 2022, on American English). More recently, however, research has also provided evidence of additional factors that contribute to variability. First, individual speakers may use different strategies to encode intonational categories (Gryllia et al., 2022, Na & Arvaniti, in press; see Cangemi et al., 2015, for an overview). Further, Gryllia et al. (2022) has also found differences in realization linked to the elicitation task, with some tasks leading to more instances or more exaggerated instances of specific tonal events.

A ubiquitous factor influencing alignment is *tonal crowding*, the presence of more tones than tone bearing units (TBUs). This is evident if one compares panels a and b in Figure 4: in *jam*, the entire tune is realized in compressed form on one TBU, while in *the marmalade*, each tonal element is realized on a different syllable and the initial *the* has no tonal specification. This mismatch between tones and TBUs that results in tonal crowding can lead to several adjustments, from retracted target alignment (e.g., Arvaniti & Ladd, 2009, on Greek), to target undershoot and the overall compression of tunes, to the elimination of some targets (*truncation*); see, e.g., Arvaniti et al. (2017) on the use of all these strategies in Polish calling contours.

In addition to tune changes, tonal crowding can lead to segmental adjustments to accommodate tunes that would otherwise have insufficient sonorant material or time to unfold (see Roettger & Grice, 2019, for an overview). Such adjustments include vowel epenthesis (Roettger & Grice, 2019, on Tashlhiyt Berber; Dell & Elmedlaoui, 2002, on Moroccan Arabic; Frota et al., 2016, on varieties of Portuguese), reduction in vowel devoicing (Beckman & Venditti, 2010, on Japanese; Andreeva & Koreman, 2003, on Bulgarian), and vowel lengthening (Arvaniti et al., 2017, on Polish; Frota, 2002, 2014, on Portuguese; Heston, 2014, on Fataluku).

These interactions, whether they lead to tune or segment adjustments or to both, indicate a close relationship between tunes and text. This view is further supported by recent evidence that tonal categories are realized using redundant cues expressed on segments, such as changes to segmental duration, amplitude and voice quality (see, e.g., Hu & Arvaniti, in press, and references therein). The close connection between tune and text runs counter to the view of intonation as a superimposed layer, and illustrates why it is essential to take intonation into consideration when studying segments. From the point of view of intonation research, however, the resultant variability poses issues for analysis, which have led to a variety of responses.

The most traditional approach has been to use AM representations as transcriptions to capture gradient phonetic differences. This is sometimes done by using both phonological and phonetic levels of AM-like annotation (e.g., Jun, 2005b, on Korean; Prieto, 2014, on Catalan), in an approach similar to the "systematic phonetic" level of analysis (i.e., the positing of discrete allophones). Discretizing gradience is also the cornerstone of several annotation systems, including IViE (Grabe, 2001/) PoLAR (Ahn et al., 2019), RaP (Dilley et al., 2006), and the International Prosodic Alphabet (Hualde & Prieto, 2016), in which specific tonal alignment configurations are related to distinct AM-like representations.

The opposite approach is to treat variability as a matter of phonetic realization. This is advocated by Beckman et al. (2005), who argue that the autosegmental representations of tunes should be abstract and examined in tandem with the rich information provided by the acoustic signal, not more fine-grained symbolic representations. A similar approach is taken by Atterer & Ladd (2004), Arvaniti et al. (2006a), and Arvaniti (2016). These authors argue that details of tonal alignment or differences in scaling between accents should not be incorporated into phonological representations as a matter of course. Rather, phonological representations should be pared-down to elements that have a contrastive function in the system under investigation (determining contrastive function requires recourse to meaning, a point we return to in §5). This approach is in line with Ladd's arguments against a level of systematic phonetics (Ladd, 2014, ch.2), which is ultimately too fine-grained for some purposes and not fine-grained enough for others (for arguments against systematic phonetics in intonation, see Arvaniti, 2016).

A related approach has been to develop measures that quantify variation in production and connect it to perception. One such measure is Tonal Center of Gravity (TCoG) proposed by Barnes et al. (2012). TCoG represents an average point in time, weighted by F0, such that TCoG values differ between domed and scooped contours, such as L+H* and L*+H, respectively. TCoG values support the auditory impression that domed accents sound higher than scooped ones, which remain low in pitch for longer (see Figure 1 for illustrations of these two accents). The ProPer toolbox (Prosodic analysis with Periodic energy, Albert, 2023; Albert et al., 2018) is a similar method that returns several syllable-based measurements reflecting both F0 features (e.g. Synchrony, a measure of the F0 trend and swiftness within a syllable, based on the TCoG), and well-known measures of acoustic salience, such as *mass*, the integral of duration and energy (for proposals and uses of this integral, see Lieberman, 1960; Beckman, 1986; Arvaniti, 1991; Turk & Sawusch, 1996; Gordon 2002).

A final approach has been to adopt articulatory phonology principles (Browman & Goldstein, 1986), and move away from the phonetics-phonology dichotomy and towards the integration of continuous and categorical aspects of intonation (cf. Grice et al., 2017). Research along these lines has modelled intonation using dynamical systems. Roessig et al. (2019) relied on predetermined AM categories to investigate rising German accents (H+!H*, H* and L+H* in GToBI; Grice et al., 1996), focusing on the F0 excursion from the pretonic syllable to the accentual H. Their attractor-based modelling captured the differences between the three accents as well as variation across and within speakers. Iskarous et al. (2024), on the other hand, did not use predetermined categories, thus addressing the criticism by Pierrehumbert & Pierrehumbert (1990) that in many dynamical models, "the discrete is specified, not derived" (cited in Iskarous et al., 2024: 3). Iskarous et al. built models in which various parameters – F0 level, velocity, peak velocity, span, and latency (the time difference between the onset of an accentual rise and its peak velocity) were incrementally added; by employing a stochastic dynamical system they also incorporated noise, thereby accounting for within-category variability. According to the authors, the properties of their final system allow for the categories of MAE pitch accents to emerge naturally, rather than being stipulated in advance.

## 4. Tame Intonation (TINT): engaging with intonational phonology

The view taken in our own work is summarized as TINT which stands for *tame intonation*, a play on Bolinger's description, popularized by Gussenhoven (2004, ch. 4), that intonation is "a half-tamed savage" (Bolinger 1978: 475). We argue that this view, which sees intonation as not quite part of linguistic structure, is the reason why intonation is often held to different standards than segments: while the sources of variation discussed in §3 have long been recognized as inevitable in speech, similar types of variability are seen as deeply problematic and undesirable in intonation (for a detailed discussion, see Arvaniti et al., 2024). To avoid this pitfall and focus on phonological structure, one strategy in our research is to seek parallels between phenomena we observe in our intonation data and well-known sources of variability in segments, before assuming that intonation needs a different approach (for examples, see Arvaniti, 2016, and Arvaniti et al., 2024).

In our view, intonation is best understood if we retain the separation of an abstract level of phonological representation coupled with a phonetic implementation module (the properties of which may well arise from dynamical systems and are learned by means of exemplars used in production and perception; Pierrehumbert, 2016). An issue that arises if this approach is adopted is how to determine the abstract elements that make up a tune, and disentangle them from inevitable variability and noise, linguistic sources of variation, and gradience related to paralinguistic and indexical factors. To achieve this goal, we have adopted a number of methods, briefly outlined below.

### 4.1 Rapid Prosody Transcription

In Rapid Prosody Transcription (RPT) proposed by Cole et al. (2010), untrained participants listen to utterances and mark on unpunctuated transcripts phrase boundaries and the words that sound prominent to them. Boundary strength and word prominence are operationalized as participant agreement; e.g., prominence is measured as the percentage of participants who marked a word as prominent (*p-score)*. Although RPT conflates metrical strength and intonation and its results are susceptible to changes in the instructions given to participants (Cole et al., 2014; Cole et al., 2019; Orrico et al., 2024), it has the advantage of requiring neither annotator training nor annotator agreement, and yields replicable results (Orrico et al., in press). The most popular use of RPT is the investigation of the linguistic correlates of prominence by examining the effect of a posteriori selected factors on p-scores or their binary equivalent (Baumann & Winter, 2018; Im et al., 2023).

In our own research, we have used RPT to address different questions. In Orrico et al. (in press), we used RPT to test a particular hypothesis about the H* vs. L+H* distinction in English: in Pierrehumbert (1980) these accents are considered phonologically distinct, and Pierrehumbert & Hirschberg (1990) relate them to different meanings (informally, H* to new and L+H* to contrastive information). Ladd, on the other hand, has argued that L+H* is not a distinct accent but an emphatic version of H* (Ladd & Morton, 1997; Ladd, 2008, ch. 3, inter alia). We explored Ladd's contention using RPT, reasoning that, if correct, all L+H*s should be rated as more prominent than H*s.

Two main results emerged from this investigation. First, participants take into consideration both the phonetic difference between the accents and their pragmatic function, but also the extent to which the two match. This indicates that speakers have clear intuitions about form-meaning mappings: our participants' responses were much clearer when accent form and meaning matched – i.e., when L+H*s were used contrastively and H*s non-contrastively – but were significantly more uncertain with mismatches (e.g., with H*s used contrastively). We return to the form-meaning mapping in §5.

Second, the relative weight of form and meaning in determining accent prominence varied based on participant musicality (measured using MiniPROMS; Zentner & Strauss, 2017), empathy (measured as Empathy Quotient; Baron-Cohen & Wheelwright, 2004), and autistic traits (measured as Autism Quotient; Baron-Cohen et al., 2001). There results, which echo earlier findings by Roy et al. (2017), and Bishop et al. (2020), highlight the role of individual variation and its root causes. Further, the fact that the differences among participants, in terms of greater attention to intonation form or meaning, were replicated in our two studies indicates that the patterns we uncovered reflect stable differences in the population as a whole (Orrico et al., in press).

In Orrico et al. (2024), we used RPT to explore the contention of AM that intonation is independent of metrical structure and that, consequently, F0 changes are not prominence lending (see §2). Greek participants responded to Greek broad and narrow focus statements (realized with H* L-L% and L+H* L-L%, respectively), wh-questions (L*+H L-H%), and polar questions (L* H-L%). The participants overwhelmingly marked as prominent the words that were metrically strong, irrespective of F0 level (e.g. L+H* and L* had the same probability of being selected), while unaccented words, such as those carrying the H-L% edge tones of polar questions, were not considered as prominent even when high in pitch These results argue against a universalist view that assumes that prominence equates to acoustic salience and highlight the role of metrical structure in interpreting tunes.

**4.2 Separating phonetic from pragmatic criteria**

In our research we have also endeavored to avoid the annotation pitfall of simultaneously weighing phonetic, pragmatic, and contextual criteria (an inevitability when annotators listen to audio while considering waveforms, spectrograms and pitch tracks). As the results of Orrico et al. (in press) discussed in §4.1 indicate, this standard practice makes it difficult to determine whether annotators respond to intonation or pragmatic information or both, and if the latter to what extent they rely on each factor. With respect to our research on English H* and L+H*, the standard annotation practices prevented us from determining whether the acknowledged annotator difficulty in distinguishing the two (see Syrdal et al., 2001, and references therein) is due to the accents forming a phonetic continuum, or to their being used interchangeably for a variety of pragmatic functions. This is not a trivial question, as positing H* and L+H* as phonologically distinct implies that the use of one instead of the other should lead to a change in meaning, as discussed in more detail in §5.

To address this issue, in Kim et al. (2024) we used independent pragmatic and phonetic in a corpus of spontaneous British English speech. High and rising accents (H* and L+H*, respectively) were categorized considering only F0-related criteria. Then, the annotated accents received an independent pragmatic label, indicating whether the accented word represented contrastive, corrective, or new information in context. These labels were determined using only written transcripts; e.g., in the dialogue in (1), *a yellow loon was* considered contrastive, independently of how the speaker produced it or how the annotator thought they would have produced it themselves.

(1) A: *I've got a wren and a wallaby*
B: *Oh! A yellow loon and a wallaby*

The separation of the phonetic and pragmatic annotations showed that while British English has two distinct phonetic accent types (falls and rise-falls, or H* and L+H*, respectively), both are used to encode new, corrective, and contrastive information (see also §4.4). This made it possible to dismiss phonetic gradience as the reason behind the disagreements and unveil the complex relation between these accents and their functions. Though this method is not suitable for all tonal contrasts, it is a useful step towards an analytical, rather than synthetic, approach to annotation.

**4.3 Functional Principal Component Analysis**

A technique that is particularly well suited for disentangling linguistically determined variation from gradience, general variability and noise, is Functional Principal Component Analysis (FPCA; Ramsay & Silverman, 2005). In FPCA, curves (for intonation analysis, F0 curves) in a given dataset are modelled in terms of the main modes (or *principal components*) in which they vary. Each curve in the dataset is then represented as a function that consists of the dataset's mean curve and a number of principal component curves (PCs) that account for the variance in the dataset. In this function, each PC is multiplied by a coefficient (or *score*) which represents the contribution of the PC to the shape of that dataset curve. (For details, see Gubian et al., 2015, Asano & Gubian, 2018, and Arvaniti et al., 2024).

As argued in Arvaniti et al. (2024), there are several advantages to FPCA, the main one being that it provides a way to understand the sources of variability in intonation. First, FPCA provides a measure of the variance accounted for by each PC, i.e. each mode in which F0 curves vary. Additionally, the PC scores can be used for follow-up statistical modeling, allowing researchers to determine the extent to which specific modes of variation present in a corpus reflect linguistic distinctions, such as two distinct accents, or other sources of variability that are of interest.

As an illustration, in Arvaniti et al. (2024), we conducted FPCA on a corpus of Greek statements produced with a H*, L+H*, or H*+L accent. FPCA showed that most of the variability in the dataset related to pitch height, reflected in PC1, which explained 65% of the variance, and pitch shape, reflected in PC2, which explained 23% of the variance. FPCA was followed by statistical modelling of the PC1 and PC2 scores, which showed that the scores for PC1 and PC2 differed by accent, confirming that the difference between H* and H*+L is largely one of height (low vs. high fall, respectively) and that between H* and L+H* one of height and shape (low fall vs. high scooped rise, respectively). Additionally, marginal and conditional $R^2$ provided an estimate of the fraction of score variance explained by the fixed effect of accent ($R^2m$ ) and the combined effect of accent and the random terms (in our analysis, tonal context and crowding, reflected in $R^2c$) (Nakagawa & Schielzeth 2013, Johnson 2014). These additional measures allowed us to better understand the overlap between the three accentual categories and separate what Bürki (2018) calls *variability* from systematic variation related to the linguistic categories.

Finally, FPCA allows for the simultaneous analysis of F0 with other signals, such as duration (Asano & Gubian, 2018. In Arvaniti et al. (2024), FPCA on the duration and F0 curves of L+H* Greek accents provided significant insights, as it showed that the two dimensions not only co-vary but enter into a trading relationship, such that less scooped L+H*s are produced with longer duration. In short, FPCA allowed us to both separate essential differences among accents from gradience and variability in their realizations and thus reach robust conclusions about the phonological distinction between H*, L+H* and H*+L in Greek, while also making it possible to aspects of phonetic realization that go beyond F0.

**4.4 Generalized Additive Mixed Models (GAMMs)**

Generalized Additive Mixed Models (GAMMs) have become a popular tool for analyzing phonetic data (Chuang et al., 2021; Sóskuthy, 2021), including intonation (Arnhold & Kyröläinen, 2017; da Silva Miranda et al., 2020; Magistro & Crocco, 2022; Steffman, 2023; Steffman et al., 2024). GAMMS are especially well-suited for intonation research because they use non-linear smoothing functions to model time-varying response variables such as F0, and capture wiggly, dynamic relationships between variables that linear models cannot handle. Further, like FPCA, GAMMs are suitable for modelling longer stretches of pitch tracks, a critical attribute as the role of F0 shape beyond tonal targets is increasingly recognized (Barnes et al., 2012; Dorokhova & D'Imperio, 2019; Joo & D'Imperio, 2023; inter alia).

A potential disadvantage of GAMMs is that they require a priori categorization of the data into predetermined AM categories. This can be avoided if GAMMs are combined with other dynamic methods, such as contour clustering (Cole et al., 2023) and FPCA. In Kim et al. (2024) we used FPCA followed by GAMM to test how the phonetic shape and pragmatic functions of H* and L+H* in Southern British English are related. The use of FPCA allowed us to determine the main dimensions in which H* and L+H* differ in British English, namely scaling (L+H* is scaled higher than H*), and shape (L+H* is rising and H* falling). At the same time, GAMM supported the PRT findings of Orrico et al. (in press), discussed in §4.1, that in British English, H* and L+H* are not used to encode distinct meanings.

In short, methods such as FPCA and GAMM have clear advantages over more traditional methodologies. First, they do not reduce the complex dynamic aspects of F0 to a handful of tonal targets, a practice that can impoverish our understanding of intonation pragmatics. Further, since they are primarily data-driven, they help curtail confirmation bias, which is inevitable during annotation. Most importantly, they reduce the dimensions along which data vary, an attribute that is especially beneficial to intonation research because it makes it possible to identify linguistically meaningful contrasts out of complex variation. Using FPCA and GAMM has made it possible to study intonation in spontaneous speech, which has often been avoided because of its inherently greater variability, thereby providing further insight into the structure and functions of intonation.

## 5. Intonational meaning

As noted in §1, intonation encodes meanings of different types that largely fall under pragmatics and include information structure (the signaling of topic, focus, and givenness; Steedman, 2007; Buring, 2016), and epistemicity (the signaling of the presence or absence of commitment to a proposition; Bartels, 1999; Gunlogson, 2008). Additionally, intonation can be used to negotiate information during an exchange; for example, it can signal the contrast between information that is consensual (agreement among interlocutors) or potentially conflictual (Steedman, 2007; Portes & Beyssade, 2015). Despite the recognized role of intonation in encoding meaning, research has often prioritized the study of form. Consequently, formal accounts of intonation meaning are relatively rare (see Pierrehumbert & Hirschberg, 1990, Steedman 2007, and Portes & Beyssade, 2015, for models, and Prieto, 2015, and Westera et al., 2021, for overviews). Below we briefly present the AM view of intonation meaning and explore the role of meaning in understanding intonational phonology.

In AM, intonation meaning does not arise directly from the F0 contour but is mediated by the structural elements that compose a tune, viz. the pitch accents and edge tones. These phonological entities are considered isomorphic to morphemes and therefore meaningful, with the meaning of each entity contributing compositionally to the intonational meaning of the utterance (Pierrehumbert & Hirschberg, 1990, Portes & Beyssade, 2015, see also Portes, 2023: 25-29). The meaning of distinct tonal events is stable but always interpreted in context and in combination with the other elements that make up the tune, as well as lexical choices, and the grammatical form of the utterance (Pierrehumbert & Hirschberg, 1990; Portes & Beyssade, 2015). As an illustration, according to Pierrehumbert & Hirschberg (1990), L* signals exclusion of the accented item from predication. This makes L* suitable for questions but the meaning of L* remains the same in statements, where it can be used to signal that the speaker believes the proposition to be incorrect or that they are stating something the hearer already knows.

The fact that meaning is computed in context entails that in AM the mapping between intonational categories and specific functions, such as questioning, focus, surprise and the like, is not one-to-one: tonal events and tunes can be used for various communicative goals which, in turn, may be achieved by different means. This observation is not an innovation of AM; it has been tacitly recognized for over a century, in accounts of British English intonation which describe the various uses and meanings of the tunes they posit (Jones, 1909, Halliday, 1970, O'Connor & Arnold, 1973, inter alia).

### 5.1 Production and perception evidence for intonational meaning and compositionality

In AM, meaning has not always been considered when positing intonation categories. For instance, the American English ToBI manuals (Beckman & Ayers-Elam, 1997; Veilleux et al., 2006) provide only phonetic criteria for determining the identity of accents (with the exception of the H* vs. L+H* distinction in Veilleux et al., 2006). Yet meaning is inevitably both a criterion in determining intonation categories in production and a key element in perception studies of intonation. Both types of evidence are reviewed below.

Production studies support both compositionality and the role of context in the interpretation of intonation meaning.  Orrico & D'imperio (2020a) showed that the meaning of Salerno Italian nuclear pitch accents, encoding speaker commitment, is retained regardless of the following boundary tone or sentence type (statement or question). Their compositional account explained the results of Orrico et al. (2019) who, as expected by the AM principles discussed above, had found that tunes do not map with statements and question in a one-to-one fashion (see also Orrico & D'Imperio, 2022). Similarly, Arvaniti et al. (to appear) provide a compositional account of the tunes used with wh-questions in Greek, but also show that their  interpretation in context can range from neutral to sarcastic, changing in the process the function of the questions from canonical to rhetorical and conjectural. Similar results are reported by Armstrong & Prieto (2015) by means of a perception experiment showing that the default tune used for questions in Puerto Rican Spanish can be interpreted as sarcastic in the right context.

Evidence for compositionality also comes from online processing: experiments conducted using the visual world and mouse tracking paradigms show that intonation is processed incrementally, with listeners making decisions about intonational meanings as soon as they hear a tonal element (inter alia, Ito & Speer, 2008, Kurumada et al., 2014, Roettger & Franke, 2019). Unsurprisingly, these decisions may be reversed once the entire tune is available. In Roettger et al. (2020), American English listeners interpreted L+H* accents as contrastive as soon as they heard them; however, tonal events downstream incompatible with the original interpretation (such as the presence of another accent) led them to re-interpret the earlier L+H*. Though Roettger et al. (2020) argue that this implies a "holistic" interpretation of the tune, it is possible to see it as a straight-forward example of cue integration, which has been widely accepted for other aspects of speech (e.g. McMurray et al., 2008), or even as an instance of garden-path. At the same time, the findings of Roettger et al. (2020) provide evidence for the importance of metrical structure in interpreting a tune: the presence of an accent downstream can lead to the reinterpretation of a L+H*, as the downstream accent entails there is a metrically stronger constituent than that accented with L+H*, and this renders the contrastive interpretation of L+H* untenable, as such an interpretation requires the deaccenting of following material; conversely, the absence of an accent downstream from a H* can lead to re-interpreting the H*-accented item as contrastive, since only contrastive accents can be followed by deaccenting.

Further, recent studies have examined the sources of individual differences in the perception of intonation and have shown that these are linked to cognitive characteristics of the participants, particularly their level of empathy (Orrico & D'Imperio, 2020b, for Italian; Esteve-Gibert et al., 2020, for French; Casillas et al., 2023, for American English). This is not surprising as individual variation due to cognitive characteristics is now reasonably well documented with respect to segments (see Yu & Zellou, 2019, for an overview). Finally, given a steadily growing body of findings indicating individual variability in the production of intonation (e.g., Lorenzen & Baumann, 2024; Hu & Arvaniti, in press;) perception studies have also focused on how listeners navigate this variability: recent evidence suggests that individuals make inferences on the form-meaning mapping based on systematic sources of variation (see Kurumada & Roettger, 2021, for an overview). For example, it has been proposed that listeners can map the same tune to different meanings based on the characteristics of the speakers, e.g. regional variety (Portes & German, 2019) or age (Warren, 2017). This process is not specific to intonational meaning but guides the processing of speech signal more generally (see Kleinschmidt, 2019 for the processing of segments).

**6. Intonation Typology**

Research on intonation typology has been hard to come by, largely because of the prevalence of the universalist view discussed in §2 (Ladd, 2001). Under this view, intonation typology is largely meaningless, as intonation is expected to be comparable across languages and driven by physiology (such as the need to take breath) and universal symbolism connecting body size and attitude with pitch height (see Gussenhoven, 2004, ch. 5, for a discussion).

Ladd (2001) contrasts the universalist view with the phonological approach, in which typological generalizations can be drawn from the observable differences in the intonational structure of different languages (see Zerbian, 2010, for similar arguments). Research in the 21st century, which increasingly includes typologically distinct and previously un(der)described languages, has shown that the universalist approach cannot explain the amply documented empirical observation that intonation systems differ substantially across languages.

These differences relate to a number of factors (see Jun, 2014b, for a proposal), some of which we discuss here. First, tonal inventories vary in size; e.g., Samoan has one accent type, while analyses of Catalan posit six (see Jun, 2014b, and references therein). Additionally, languages may differ in terms of the combinatorial possibilities of their inventory. For instance, while Pierrehumbert (1980) assumes that accents can be freely combined with each other and all edge tones, in practice some combinations are very rare (Dainora, 2006). In some systems, particular combinations may become consolidated into tunes with a specific meaning; e.g., in Greek, a default L* H-L% tune with specific alignment properties (Arvaniti et al., 2006b) is used exclusively for polar questions; in contrast, there is no default tune for polar questions in English. Further, languages vary with respect to the location of the *nuclear pitch accent* (typically the final accent of an utterance, associated with the metrically strongest constituent and sometimes referred to as *sentence stress*). English and Greek wh-questions have comparable syntactic structure, but in English the nuclear accent is on the last word, while in Greek it is on the fronted wh-word (Arvaniti & Ladd, 2009). The frequency of accentuation is another typological difference: in English some content words may remain unaccented (e.g. the verb *won* in Figure 2), while in Spanish and Greek all content words are typically accented (Prieto & Roseano, 2018; Arvaniti & Baltazani, 2005). Finally, the functions of intonation and the means by which they are achieved may also vary, with intonation having primarily a culminative or a demarcative function (Jun, 2014b). The former type is exemplified by Germanic languages which use intonation to encode information structure by means of accentuation and subsequent deaccenting. The latter type is exemplified by Korean and Japanese, in which the same effect is achieved by means of *dephrasing*, i.e., the grouping of separate APs into one AP (Venditti et al., 1996). Yet, not all languages fit this dichotomy equally well; in some Romance languages, such as Italian, deaccenting is at best a limited option while dephrasing for information structure marking not possible at all (see Ladd, 2008, chap.6, for a discussion).

There is less work on a possible phonetic typology of intonation, as there is less agreement about intonation's building blocks, since analyses (and languages) carve up pitch contours in different ways and may follow different traditions (see Ladd, ch. 4, for a discussion). H* and L+H* are again a case in point: Jones (1909) proposed both rising-falling and falling accents for British English (equivalent to AM L+H* and H*, respectively), but he later adopted the proposal of Armstrong & Ward (1926) who posited only rises and falls (Jones, 1932). The rise vs. fall distinction of Armstrong & Ward became a cornerstone of the British School of intonation, such that recent analyses of (British) English intonation treat falls as essential (Gussenhoven, 2004, 2016), while comparable falls are seen as incidental in AM analyses based on American English. Although, as the work of Kim et al. (2024) indicates, there is an empirical base for the differences, decisions about how to carve the F0 contour into units remains a challenge. Gussenhoven (2016) convincingly argues that the descriptive adequacy of an analysis must be scrutinized to determine whether it over- or under-generates tunes. However, until extensive corpora of spontaneous speech are analyzed using data-driven methods, this practice is unlikely to take hold.

Because of these challenges, typological distinctions regarding intonation phonetics tend to be smaller in scale. Grabe et al. (2000) argued that some language varieties truncate and others compress in the presence of tonal crowding. Though this purported distinction is often readily accepted, it is not supported by empirical evidence which shows that many languages use all available strategies, depending on the tune (Ladd, 2008, ch.5; Arvaniti et al. 2017).[4] Arvaniti (2022b) proposed that languages differ in their association strategies: in languages showing stricter tonal alignment (such as Greek; Arvaniti et al., 1998), association percolates from a metrical head, such as an ip, to a specific tone bearing unit (TBU), such as the head of a foot; in languages with looser alignment, such as Serbian and Croatian (Smiljanić, 2009) and Tashlhiyt Berber (Roettger et al., 2013), percolation does not take place. Although such proposals have limited appeal, they go some way towards explaining systematic differences in the realization of intonation.

## 8. Conclusions

In summary, the evidence reviewed above supports the AM understanding of intonational phonology: tunes consist of independent tonal events that phonologically associate with phrasal boundaries and metrical heads (in the languages that mark them). The phonological elements that make up tunes encode primarily semantic-pragmatic meaning, which is compositional in nature and interpreted in light of its linguistic and pragmatic context. Most importantly, the main tenet of AM, namely that intonation is a phonological subsystem is empirically supported, in that much of the variability encountered in pitch contours and their meaning can be understood if we move away from the scrutiny of phonetic detail to tune abstractions.

[4] Based on their findings from Polish, Arvaniti et al. (2017) have proposed instead that the solution of tonal crowding be used as criterion for tonehood, with truncation indicating the lack of an underlying tone and compression indicating its presence.

The above should not be taken to mean that investigating the phonetics of intonation is redundant. Indeed, the research of the last two decades has shown that we need a richer understanding of intonation phonetics and that achieving this understanding can lead to insights about phonology. While the temporal alignment of tonal targets to text, as originally posited by AM, is important, recent evidence suggests that intonation phonetics is richer than the investigation of tonal targets suggests: first, the course of F0 between tonal targets can be critical, and second, tonal events are redundantly realized by means of multiple cues that affect the segments with which tonal events are co-produced. In turn these findings indicate that it is high time we abandoned ill-conceived metaphors of intonation as peripheral or "suprasegmental", as its interactions with segments are pervasive and its role essential in understanding both aspects of grammar, such as information structure and its expression, and communication.

We have discussed several findings that would benefit from seriously engaging with the phonological nature of intonation. Such an approach would avoid treating as exceptional findings that are commonplace in the study of segmental phonetics and general processing. Perhaps the most important aspect of intonation that needs such a fresh approach is variability. Variability is often treated as proof of intonation's exceptionality, but is in fact typical of speech, as we have shown, provided that methods suitable for the investigation of intonation (including, but not limited to, GAMM and FPCA) are adopted, instead of assuming that standard visual representations, such as pitch tracks, are sufficient for intonation research.

In this endeavor, we believe that the role of meaning is critical, though it has often remained limited in intonation research. Formal models of meaning can provide robust criteria for production and can guide research in perception, provided the meanings investigated are broad enough to account for the uses of intonational elements across contexts (e.g., presence vs. absence of commitment rather than statements vs. questions).

Finally, recent studies, including our own research, show that in order to achieve a better understanding of intonational phonology it is important to explore typologically varied languages, spontaneous data in varied styles, and use diverse research paradigms, while always keeping in mind possible parallels with the study of segmental phonetics and phonology. These strategies should lead to more robust findings, and also to a better understanding of intonational typology and the contribution of intonation in communication.

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